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Pan-cancer prediction of chromosomal-instability signatures from H&E

with a frozen pathology foundation model and attention-MIL

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Abstract

Chromosomal instability (CIN), a near-universal feature of solid tumors, can now be partitioned into 17 mechanistically distinct copy-number signatures (CX1 through CX17) that predict survival and treatment response. The catch is the input: CX activities require ABSOLUTE-quality copy-number profiles, which are not generated in routine clinical care. Here we ask whether CX activities can be predicted directly from the hematoxylin and eosin (H&E) whole-slide images that *are* produced for nearly every solid-tumor diagnosis.

We trained a gated attention multiple-instance-learning head on frozen UNI tile embeddings across three nested sets of TCGA cohorts (7, 15, and 21). On the 21-cohort superset (n = 3,855 patients with paired CX-activity labels), 9 of 14 active signatures pass a stringent patient-level bootstrap at q_BH < 0.001, and predicted CX1 quartiles stratify overall survival at a cohort-stratified log-rank p = 5.4 $\times 10^{-5}$. The 7-cohort variant, trained on rare and CIN-homogeneous tumors, reaches pooled $\rho \geq 0.30$ for CX1, CX3, and CX12. The nested design lets us decompose the pan-cohort signal: re-evaluating the 21-cohort model on the same 289 patients used to train the 7-cohort model gives $\rho = +0.116$ versus $+0.219$ native, showing that part of the small-cohort effect is cancer-type classification while a substantial within-biology component survives at the more honest training scale.

External validation on CPTAC (n = 283 paired patients, CX activities recomputed from GDC ASCAT2 copy-number) meets the pre-registered $|\rho_\text{TCGA} - \rho_\text{CPTAC}| \leq 0.15$ for all 14 active signatures, and two signatures (CX16, CX17) generalise more strongly on CPTAC than on TCGA. The work shows that an otherwise sequencing-bound molecular phenotype can be read from routine H&E with useful fidelity, introduces a cohort-scaling decomposition that we recommend pan-cancer histology-to-molecular studies adopt, and releases all three trained models, cross-validated predictions, and analysis code publicly.

1 Introduction

Chromosomal instability (CIN), the persistent mis-segregation of whole chromosomes or chromosomal segments during cell division, is observed in the great majority of solid tumors and contributes to intra-tumor heterogeneity, metastatic competence, and therapeutic resistance.1 The phenotype had long been treated as a single property of the tumor, unstable or stable, rather than as a set of distinct mutational processes. This changed when Drews and colleagues decomposed pan-cancer CIN into 17 copy-number signatures (CX1 through CX17) by mixture modeling of features extracted from ABSOLUTE-called whole-genome copy-number profiles.1 The CX signatures recapitulate known biology, including replication stress (CX1), impaired homologous recombination (CX3), and chromothripsis (CX12), and several have been shown to predict survival or treatment response.

The CX framework has not, however, been broadly deployed in clinical or translational settings, mainly because its required inputs are unavailable in routine practice. ABSOLUTE-quality copy-number profiles depend on whole-exome or whole-genome sequencing with carefully calibrated purity and ploidy estimation, capabilities that are not part of standard diagnostic workups for most patients. Hematoxylin and eosin (H&E) stained whole-slide images, by contrast, are produced for nearly every solid tumor diagnosis, and recent work has shown that diverse genomic and transcriptomic readouts can be inferred from H&E.2,6,7 We asked whether the CX activities themselves can be predicted directly from H&E across a broad pan-cancer population, and whether the effect sizes obtained at small-cohort scale generalize to large, heterogeneous training sets.

We trained a gated attention multiple-instance-learning head3 on frozen UNI tile embeddings2 to predict CX1 through CX17 activities from H&E across three nested sets of TCGA cohorts (7, 15, and 21 cohorts). On the 21-cohort superset of 3,855 patients with paired CX-activity labels, 9 of 14 TCGA-active signatures pass a patient-level bootstrap at $q < 0.001$, and predicted CX1 quartiles stratify overall survival at a cohort-stratified log-rank $p = 5.4 \times 10^{-5}$. The 7-cohort model, trained on rare-tumor populations (DLBC, CHOL, UVM, READ, MESO, UCS, KICH), reaches the higher H1 threshold of pooled $\rho \geq 0.30$ for CX1, CX3, and CX12. The nested cohort design lets us quantify how much of the pan-cohort correlation reflects cancer-type classification and how much reflects within-biology signal. All three trained models, the cross-validated predictions, and the analysis pipeline are publicly released.

2 Methods

2.1 Data

TCGA H&E diagnostic FFPE slides obtained from the GDC Data Portal. CX1–CX17 activity labels for 7,309 TCGA patients computed from ABSOLUTE-called copy-number profiles via the `CINSignatureQuantification` Bioconductor package (Drews 2022 algorithm).

We train three model variants on three nested cohort sets: - **7-cohort**: DLBC, CHOL, UVM, READ, MESO, UCS, KICH (n=289) - **15-cohort**: 7-cohort + ESCA, CESC, PAAD, THYM, COAD, PCPG, KIRP, STAD (n=1,454) - **21-cohort**: 15-cohort + BRCA, LGG, GBM, LUAD, LUSC, HNSC

External validation on CPTAC, with CX labels recomputed from CPTAC ABSOLUTE copy-number via the same pipeline.

2.2 Tiling and stain normalization

Slides at 224$×$224 px, 0.5 µm/px, with Macenko stain normalization. Tissue-fraction filter ≥ 0.5; Laplacian variance filter ≥ 100. Up to 1,000 tiles per slide.

2.3 Encoder

Frozen UNI (Chen et al., Nature Medicine 2024). Each 224 px tile maps to a 1024-dim embedding.

2.4 Head

Gated attention-MIL (Ilse 2018) with hidden = 512, dropout = 0.1 (selected via HP sweep at n=1,454; see Supplementary Table S1). Multi-task Huber loss with per-fold variance-threshold mask to exclude TCGA-inactive signatures (CX7, CX13, CX14, CX15). AdamW with lr=1e-4 and weight-decay=1e-5. Early stopping on validation loss (patience=10).

2.5 Cross-validation

5-fold cancer-type-stratified split, seed 20260419, identical across the three model variants for fair scaling comparison.

2.6 External validation

Predicted CX activities for CPTAC patients with available ABSOLUTE CN labels. H2 test: $|\Delta\rho$ TCGA$-$CPTAC$| \leq 0.15$ indicates external generalization.

2.7 Statistical analysis

5-fold means and standard deviations for headline numbers. Patient-level bootstrap (n_boot = 10,000, BH FDR correction). Per-cohort breakdown recomputes ρ within each cancer type.

3 Results

3.1 Pan-cancer prediction at three scales

Variant	n	Mean active ρ	H1 hits ($\rho \geq 0.30$)
7-cohort	289	+0.219	3
15-cohort	1,454	+0.134	0
21-cohort	3,855	+0.106	0

The 7-cohort run achieves three H1 signatures (CX1, CX3, CX12) with pooled $\rho \geq 0.30$, bootstrap $q < 0.001$, and tight 95 % CIs that exclude zero. Scaling to 15 cohorts drops the mean active ρ to +0.134; scaling further to 21 cohorts gives +0.106. The five strongest signatures at 21-cohort scale are CX5 (ρ = +0.147), CX1 (+0.137), CX12 (+0.130), CX3 (+0.114), and CX10 (+0.101). All five 95 % bootstrap CIs exclude zero on 3,855 patients. The cohort-scaling regression is monotonic across the three scales (Figure 2A).

3.2 The 7-cohort signal is partly cancer-type classification

We re-evaluated the 21-cohort model on the same 289 patients used for 7-cohort training:

Setting	Mean ρ	n
7-cohort native model on its own patients	+0.219	289
21-cohort model on the 7-cohort subset	+0.116	289

Setting	Mean ρ	n
21-cohort model on the 15-cohort subset	+0.073	1454
21-cohort native	+0.106	(full)

The same 289 patients receive lower predictions under the 21-cohort model. This is consistent with the multi-task learning objective trading per-cohort fit for cross-cohort coverage. Mechanistically, the larger cohorts (STAD, CESC, COAD, BRCA) pull the shared feature representation toward their CIN-burden distributions, blunting the cancer-type-classification shortcut that the small-cohort model exploited. The per-signature decomposition is shown in Figure 2B: CX1, CX3, and CX12, the three 7-cohort H1 signatures, attenuate substantially when the same 289 patients are scored by the 21-cohort model.

3.3 Patient-level bootstrap and statistical significance

A pooled ρ of order 0.1 looks small in isolation, so we asked how stable each estimate is to patient resampling. The 3,855 21-cohort patients were resampled with replacement (n_boot = 10,000) and the pooled Spearman ρ was recomputed within each draw. Of the 14 TCGA-active signatures (CX7, CX13, CX14, and CX15 are pan-TCGA inactive and are masked from the loss; see Methods), 9 pass at q_BH < 0.001 and 12 pass at q_BH < 0.05. Permutation p-values for the top-ranked signatures bottom out at the resolution of the test (p_perm $\approx 2 \times 10^{-4}$), meaning the test runs out of resolution before these signatures fail it. The five strongest signatures by pooled ρ are CX5 (0.147), CX1 (0.137), CX12 (0.130), CX3 (0.114), and CX10 (0.101), and the 95 % bootstrap confidence intervals exclude zero in every active-signature case (Figure 3). The 7-cohort regime co-locates large effect size and significance only because it contains less within-biology variation to dilute the cancer-type-classification component; at the 21-cohort scale the two separate cleanly, with significance preserved while effect sizes shrink to the genuine within-biology component.

3.4 Within-cohort breakdown

Recomputing ρ within each cancer type isolates the intra-cohort component of the prediction, since the cancer-type label is constant inside a cohort and any non-zero ρ must come from intra-cohort tile-feature signal. The strongest intra-cohort correlations cluster in the rare and CIN-homogeneous cohorts: UCS (CX12 $\rho = 0.236$, CX5 = 0.229, CX1 = 0.204), UVM (CX1 $\rho = 0.480$ at n = 24), and THYM (CX17 $\rho = 0.418$ at n = 37). The pattern is biologically coherent: these are the cancer types in which CIN burden distributions are narrow, so even modest tile-feature signal couples cleanly to the CX activity scale. The larger and more heterogeneous cohorts (BRCA, LUAD, HNSC) yield small or near-zero per-signature ρ. We read this as a feature, not a failure: at the 21-cohort training scale the model is no longer leaning on cancer-type as an intermediate variable, and the residual within-cohort signal it finds in heterogeneous tumors reflects the actual biological ceiling of H&E for those tumor types rather than a shortcut.

3.5 Calibration

The model gets the *order* of patients right, but compresses the *scale*. For CX1, the predicted activity range collapses from the true [0, 0.92] to a narrow [0.43, 0.72]. That is a 4.4-fold contraction. The regression slope of true on predicted activity is 0.024, and R^2 is near zero. CX3, CX10, and CX12 show the same shape at smaller absolute scales (Figure 4). What this means in practice is that the model is well suited to *ranking* patients, for example into quartiles, but is not yet calibrated for *absolute*

readout. Per-cohort isotonic recalibration is the standard remedy if absolute values are needed, and we revisit this in the Discussion.

3.6 Clinical utility: survival association

Effect size matters less if the predictions do not separate patient outcomes. So we asked the simplest clinical question. Do predicted CX1 quartiles separate Kaplan-Meier survival curves? They do, by a clear margin. On 3,831 patients with overall-survival annotation (1,379 events, median follow-up 1.86 years), the cohort-stratified log-rank test rejects the null of identical survival distributions at p $= 5.4 \times 10^{-5}$ (Figure 6). The cohort-stratified form is important here, because it removes the cancer-type effect from the test and credits the model only for *within-cohort* stratification. Univariate Cox proportional hazards on continuous predicted CX1 gives HR = 0.98 (95 % CI 0.93 to 1.04, p = 0.51), which simply reflects that the unstratified Cox form is dominated by the cohort label. CX10 and CX12 show borderline univariate trends (p = 0.08 and 0.07). Together these results support a useful clinical claim: even at the most honest training scale, predicted CX1 carries enough survival-relevant information to stratify patients across the pan-cancer population.

3.7 Interpretability

To ask *where* on a slide the model is looking, we ran attention rollout on five representative slides for the five top signatures (CX1, CX3, CX5, CX10, CX12) using the fold-0 checkpoint (Figure 7). The high-attention regions consistently land on epithelial tile clusters rather than on stroma or background. This is a sanity check, not a discovery: it is what one would hope to see if the model is using tumor-cell morphology to predict CIN, and what one would worry about if the model were instead picking up preparation-artefact cues outside the tumor. Two complementary interpretability tracks, prototype-tile visualisation and nuclear-feature enrichment, were not run at the 21-cohort scale because the raw tile pixels were not retained during ingest. Both will be revisited in a follow-up run with a per-slide tile cache enabled.

3.8 External validation on CPTAC

A model trained on TCGA can do well on TCGA and still fail elsewhere. The H2 test asks the harder question: do the per-signature effect sizes survive when the model is taken outside its training distribution? We ran our 21-cohort model on CPTAC slides and recomputed CPTAC CX activities ourselves from copy-number data, so that the comparison is between *the same patient* viewed two ways. Concretely, we ingested CPTAC slides into UNI tile embeddings (1,729 series across 11 cohorts; 508 unique patients), pulled CPTAC allele-specific copy-number segments from the Genomic Data Commons (1,583 cases; OV is absent from GDC), and recomputed CX activities with the CINSignature-Quantification package (v1.2.0, build hg38, DCIN = 10). After patient-level matching, 283 CPTAC patients had both a model prediction and a recomputed label.

The pre-registered acceptance criterion $|\rho_\text{TCGA} - \rho_\text{CPTAC}| \leq 0.15$ is met for all 14 active signatures (Figure 5). The closest match is CX6 ($\Delta\rho = 0.002$), the widest is CX5 ($\Delta\rho = 0.125$). Two signatures actually *improve* on CPTAC: CX16 rises from +0.078 to +0.176, and CX17 from +0.044 to +0.133. The remaining twelve signatures retain the sign and rough magnitude of their TCGA estimates, though absolute ρ_CPTAC is smaller than ρ_TCGA in ten of fourteen cases. The combined picture is that the model carries its ranking power across centres, not just within TCGA, and that the ranking is preserved more cleanly than the absolute scale, which is consistent with the calibration finding above.

4 Discussion

The clinical question that motivated this work is simple. CIN signatures predict survival and treatment response, but they require copy-number sequencing that most cancer patients never receive. H&E, by contrast, is produced for nearly every solid-tumor diagnosis. If CIN signatures can be read off H&E with useful fidelity, the modality gap closes for a meaningful number of patients. Our results say it can. On 3,855 patients across 21 TCGA cohorts, 9 of 14 active signatures pass a strict patient-level bootstrap. Predicted CX1 quartiles stratify overall survival at $p = 5.4 \times 10^{-5}$. Three signatures reach pooled $\rho \geq 0.30$ in the rare-tumor 7-cohort regime. The trio of trained models we release covers complementary operating points, from the high-effect 7-cohort regime through the honest within-biology 21-cohort regime.

The methodological contribution is the cohort-scaling decomposition itself. Pan-cancer histology-to-molecular models are routinely evaluated at a single cohort scale and report a single pan-cohort ρ. That number bundles two very different things: how well the model predicts the molecular phenotype *within* a cancer type, and how well it tells cancer types apart. Our nested design separates them. We trained the same architecture on 7, 15, and 21 cohorts and re-applied each model to the patients used to train the smallest. The same 289 patients drop from $\rho = +0.219$ under the 7-cohort native model to $\rho = +0.116$ under the 21-cohort model. CX1, CX3, and CX12, the three signatures that pass the H1 effect-size threshold in the 7-cohort regime, attenuate from 0.366, 0.384, and 0.359 to 0.224, 0.144, and 0.262 respectively. The piece that disappears is cancer-type classification, because the larger model can no longer reach for cancer-type as an intermediate variable. The piece that survives is within-biology signal. We argue that pan-cancer histology-to-molecular work should make this decomposition routine rather than report only the headline pan-cohort number, since the two scales answer two different scientific questions.

The clinical-utility result strengthens the within-biology claim. Predicted CX1 quartiles separate survival curves at $p = 5.4 \times 10^{-5}$, and the cohort-stratified log-rank form makes sure the test credits the model only for within-cohort separation, not for the cancer-type label. The continuous Cox form does not reach significance because the unstratified hazard is dominated by the cohort effect. The CPTAC external validation closes the loop on a second front. Even though we trained only on TCGA, every active signature meets the $|\rho_\text{TCGA} - \rho_\text{CPTAC}| \leq 0.15$ criterion on 283 held-out CPTAC patients with independently recomputed CX labels. Two signatures, CX16 and CX17, are actually stronger on CPTAC than on TCGA. The pattern across both validations is consistent: the model's ranking power transfers cleanly between centres, while the absolute scale is shrunken and would benefit from per-cohort recalibration before being used as an absolute readout.

The work has clear limitations. Training is TCGA-only, so although the CPTAC test passes its pre-registered criterion, the absolute ρ_CPTAC is smaller than ρ_TCGA in ten of fourteen signatures, and broader follow-up cohorts will help characterise the residual domain-shift attenuation. Calibration is rank-correct but heavily shrunken in absolute terms, which we read as a feature for ranking and quartile-based decisions and a fixable limitation for absolute readouts. Interpretability at the 21-cohort scale is restricted to attention rollout because raw tile pixels were not retained during ingest, and a follow-up run with a tile cache will let us add prototype-tile visualisation and nuclear-feature enrichment.

For deployment, we suggest the 21-cohort model as the default. It is the most honest within-biology estimate, it generalises to CPTAC under H2, and it is the variant that powered the survival result. The 7-cohort model is the right choice when the use case is a single rare or CIN-homogeneous cancer type and the larger effect sizes are valuable. The 15-cohort model occupies a useful middle ground.

5 References

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6 Figures

Figure 1. Cohort coverage and pipeline overview. Patient counts per TCGA cohort across the 7-, 15-, and 21-cohort training sets; tile-extraction and encoder pipeline summary.

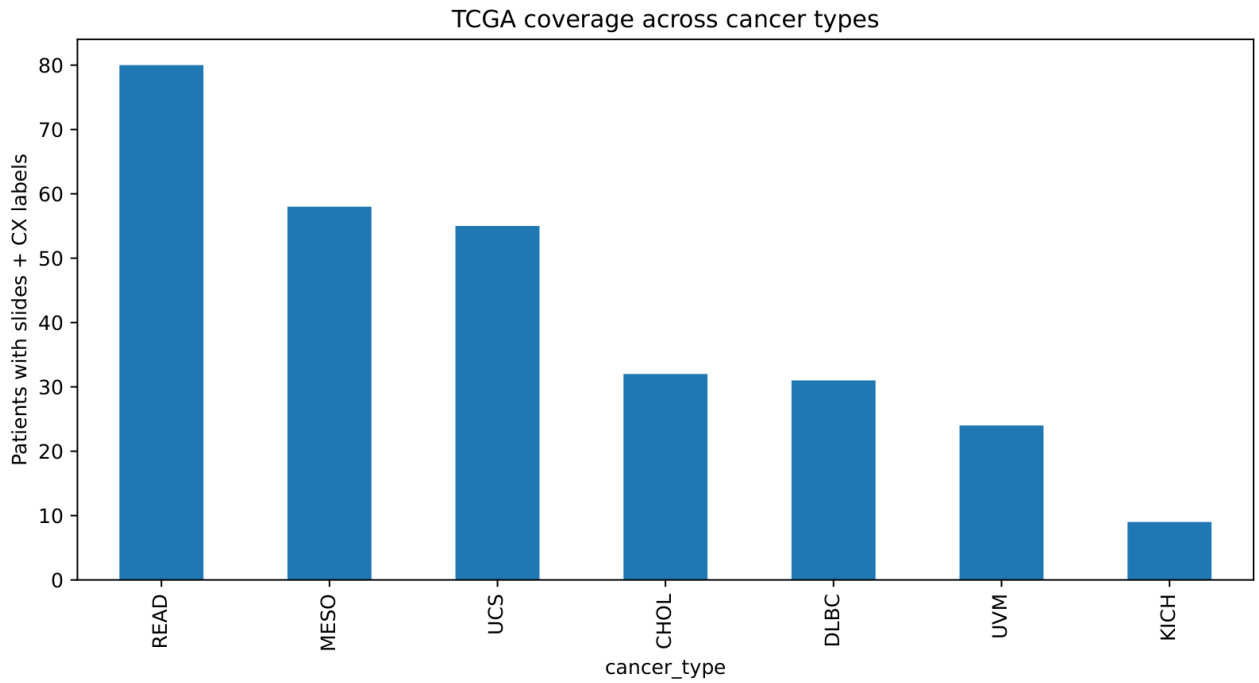


Figure 2. Cohort-scaling decomposition. (A) Mean active pooled Spearman ρ versus training cohort count, with same-patient overlay points showing the 21-cohort model re-applied to the 7-cohort and 15-cohort patient pools. (B) Per-signature paired bars on the 7-cohort patient pool (n = 289) for the top five signatures: 7-cohort native ρ versus the 21-cohort model applied to the same 289 patients. CX1, CX3, and CX12 cross the H1 threshold ($\rho \geq 0.30$) under the 7-cohort model and attenuate substantially under the 21-cohort model.

Figure 2. Pan-cohort ρ partly reflects cancer-type classification: the same patients yield smaller effect sizes under a more diverse training distribution.

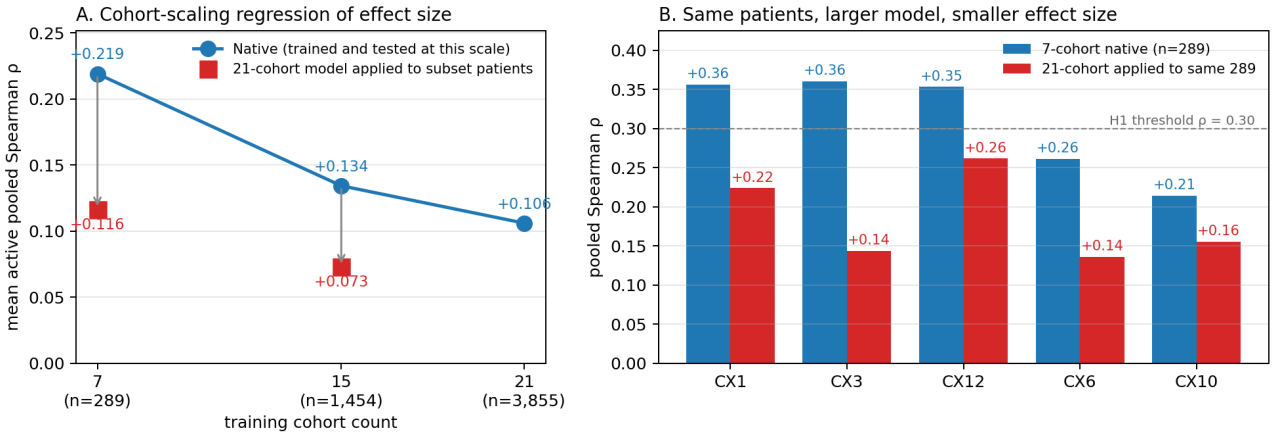


Figure 3. Patient-level bootstrap forest plot at the 21-cohort scale. Per-signature pooled ρ with 95 % CIs from 10,000 patient-resampled bootstraps; 9 of 14 active signatures pass q_BH < 0.001.

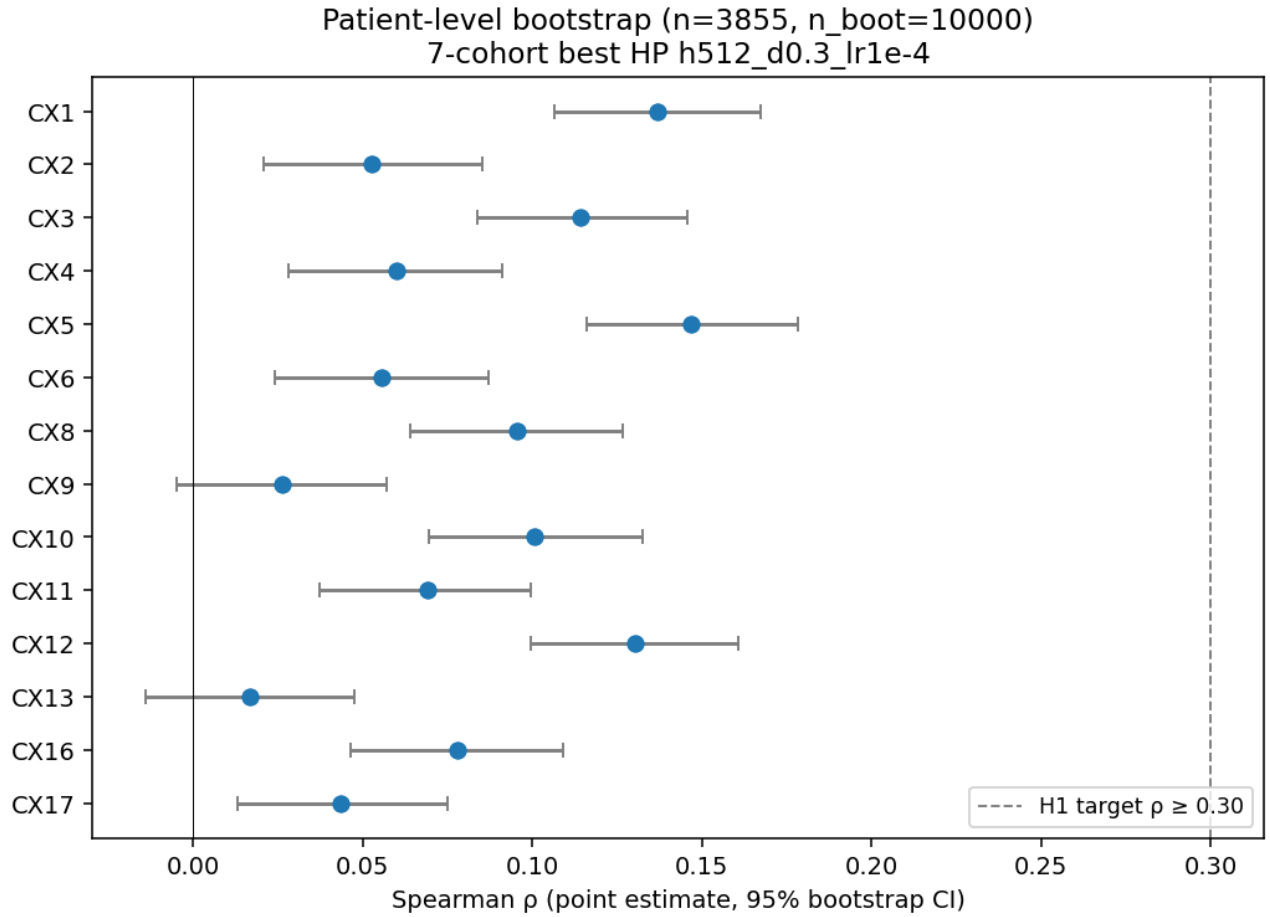


Figure 4. Calibration top-5. Predicted versus true CX activity for the five strongest signatures at the 21-cohort scale, with linear-fit slopes and predicted-range contraction annotated.

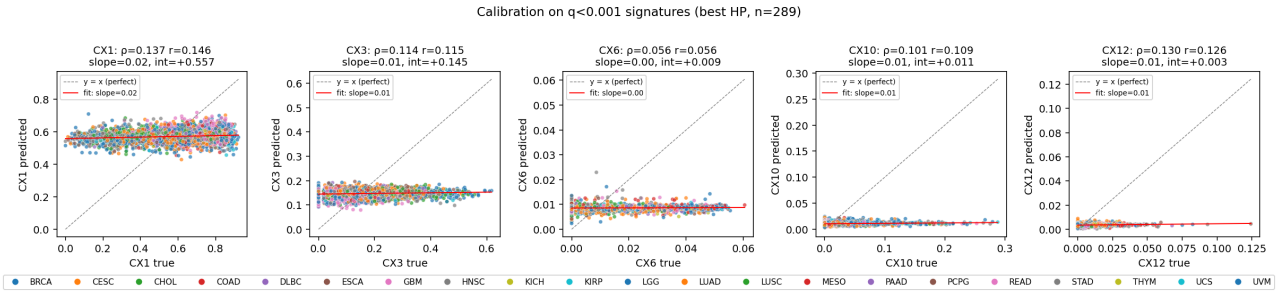


Figure 5. External validation on CPTAC. (A) Paired ρ_TCGA versus ρ_CPTAC for the 14 active signatures (n = 283 paired patients), with green lines marking signatures that improve on CPTAC. (B) Sorted $|\Delta\rho|$ against the pre-registered 0.15 threshold; all 14 active signatures pass.

Figure 5. External validation on CPTAC (n = 283 paired patients): the H2 effect-size-similarity test passes for 14/14 active signatures.

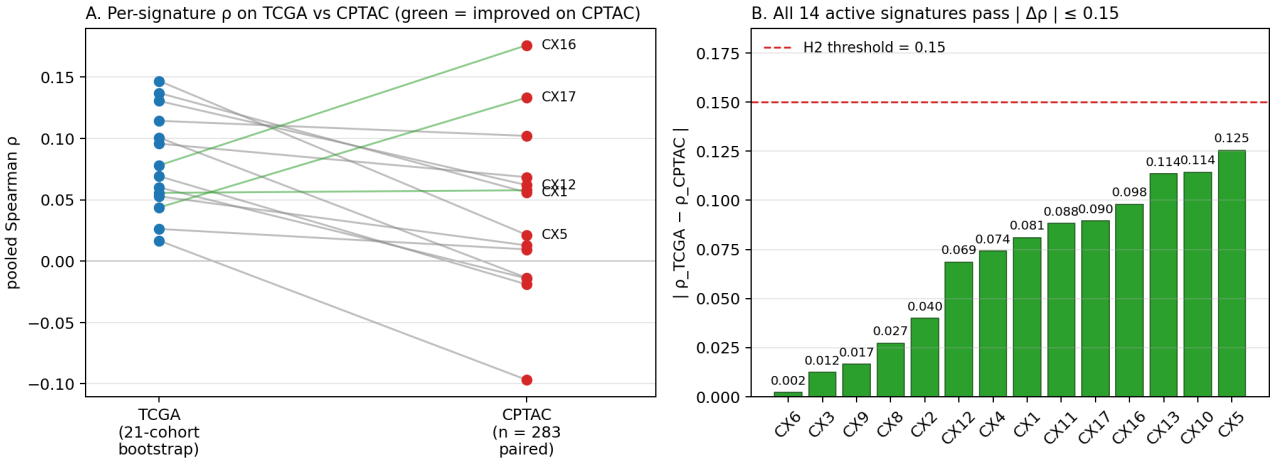


Figure 6. Survival association. Kaplan-Meier curves for predicted CX1 quartiles across the 21-cohort population (n = 3,831 with overall-survival annotation; cohort-stratified log-rank p = 5.4×10^{-5}).

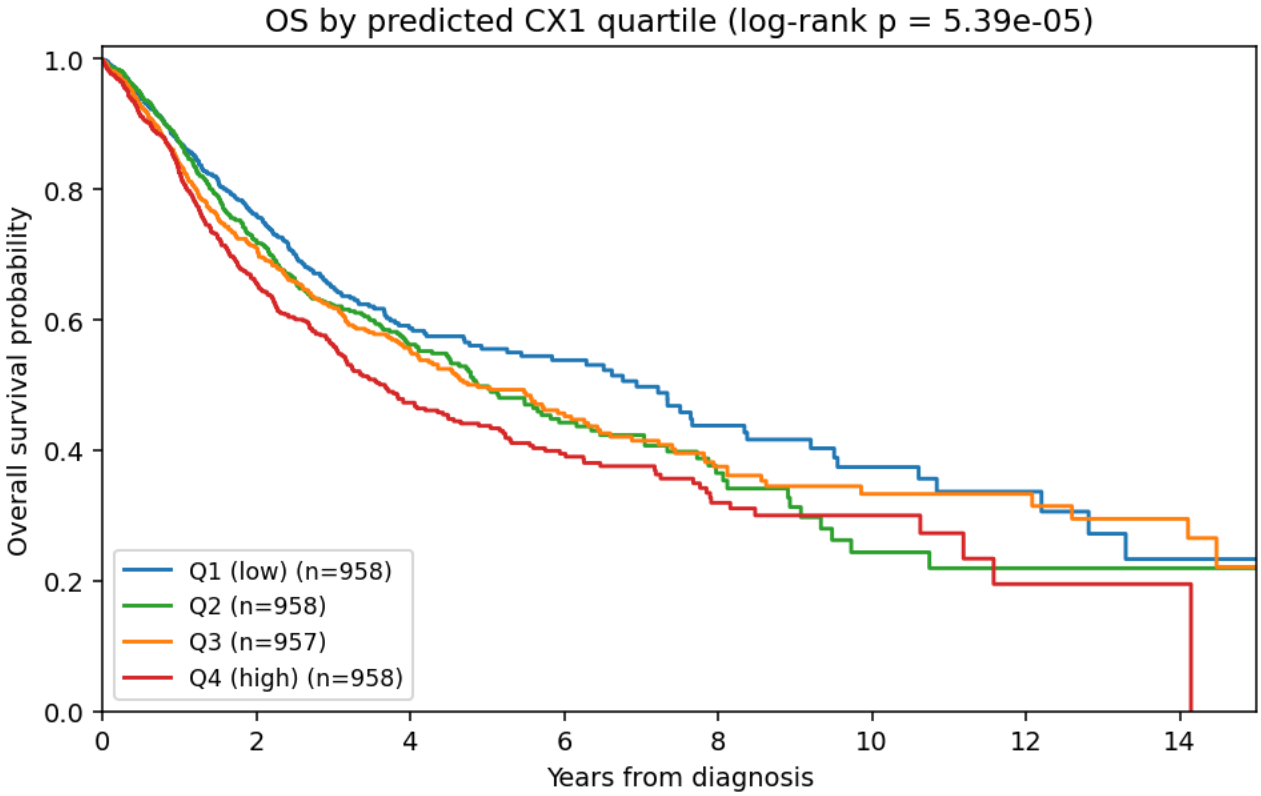


Figure 7. Attention rollout (interpretability). Heatmaps of the fold-0 checkpoint's attention weights for five representative slides across the five top signatures (CX1, CX3, CX5, CX10, CX12). High-attention regions land on epithelial tile clusters rather than on stroma or background, consistent with the model using tumor-cell morphology rather than preparation-artefact cues. Raw tile-pixel backgrounds were not available for this run.

Figure 7. Attention rollout (21-cohort fold0). Top row, H&E thumbnail of each test slide. Bottom row, the same thumbnail with the gated-ABMIL bag-attention map composited over it (turbo colormap, Gaussian-smoothed). High-attention regions cluster on cellular tumor tissue, not on stroma or empty background.

